

A Pipeline of XR-based Moving Trajectory Editing System for Choreography Production

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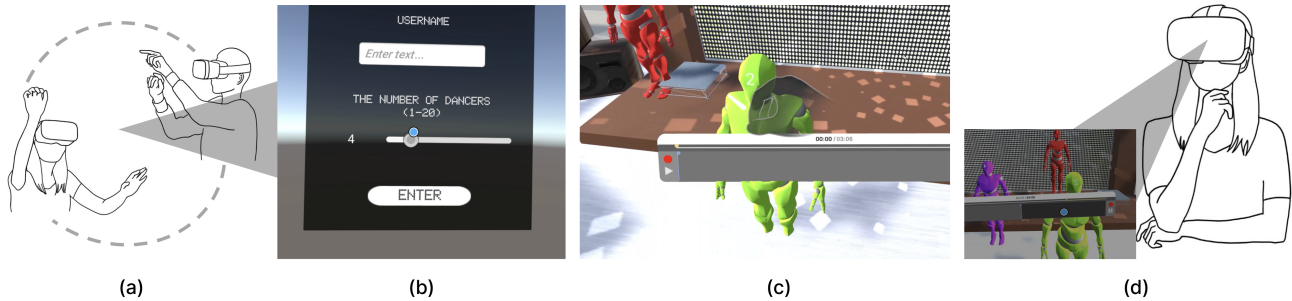


Figure 1: The UX Scenario of our system. (a) Choreographers can enter the VR environment simultaneously, (b) decide the number of dancers together through the server, (c) edit each dancer's movements, (d) and confirm the final choreography drafts of whole dancer avatars which is the result of collaboration. All choreographers participating in this production pipeline can produce overall choreography trajectories while discussing each other.

ABSTRACT

Digital twins (DT) and Extended reality (XR) are widely explored fields of study in many applications. However, the implemented application or defined framework for choreography editing is not studied, in spite of its possible increase in utility. In this work, we propose a novel approach to utilize the DT+XR environment for choreography production. Our method will contain a user interface, trajectory editing, and server in a single framework. Furthermore, studies on each technological component will be contained for detailed implementation. Unlike existing trajectory production applications, it is expected to be useful to choreographers as it can be observed from various perspectives since this system utilizes 3D space which makes a significant contribution to the choreography field by intuitively and easily creating movement lines. The movement data of each dancer saved from the user also can be augmented in various other real-world stages later to simulate the dancers' movements in advance.

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KEYWORDS

Digital Twins, Extended Reality, User Interface, User Experience, Choreography Production

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1 INTRODUCTION

Dance, a timeless form of artistic expression, has woven itself intricately through the tapestry of human history. From ancient rituals to contemporary performances, it has been a mirror reflecting the soul of cultures, emotions, and narratives. As a universal language transcending barriers, dance has been an integral part of human existence.

For the consistency and reproducibility of the work, organizing choreography has been important. Nowadays, choreography is important not only for reproducibility, but also for visual aesthetics, safety and avoidance of collisions, artistic expression and narrative, efficiency in rehearsals, and adaptability and modification. For these reasons, organizing choreography, especially planning the trajectories of dancers has been an important part of the dance industry so far.

To plan such trajectories of dancers, the conventional method is to draw the trajectories of dancers by hand on paper, which makes

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it hard to monitor the results in a real-time manner during the pre-production process. To resolve such issues, trajectory editing tools have been developed. These approaches have been developed mainly as 2D applications, which makes it hard to monitor the results from different viewpoints, being unable to consider visual aesthetics during the pre-production. And since the usage of such tools was not intuitive and hard to use, professional choreographers do not use them or even do not know their existence.

Our work is inspired by the recent advancements in extended reality (XR) and digital twin (DT) technology. XR is an umbrella term encapsulating augmented reality (AR), virtual reality (VR), mixed reality (MR), and everything in between. XR has advantages in integrating virtual and real components together via immersive user experience. XR is often considered to be related to digital twin (DT) technology. DT is a complete virtual copy of the real world, which makes it able to have real-world simulations in virtual environments. Users can have much more enveloping XR experiences in a DT environment, and such experiences are helpful in most cases. According to the benefits, there have been many attempts to utilize DT and XR simultaneously to improve conventional industries. Nevertheless, there were no specific cases in which such technologies were adopted in the field of choreography editing.

In this paper, we propose a novel approach to utilize XR environment and DT for choreography editing. Our work aims to resolve three following challenges of choreography editing: 1) Difficulty conveying clear routes of movements to each dancer, 2) Unintentional modification of formation due to various types of stages, and 3) Asynchronous collaboration with other producers. By utilizing the property of well-reconstructed 3D space that reflects the actual concert hall environment, choreographers can instantly monitor the results of their work during the pre-production stage, without having actual dancers. Using XR takes advantage in enabling the consideration of multiple viewpoints. This enables choreographers to plan and monitor their works in detail, thus it reduces the need for iterative adjustment with actual dancers in the real world, which is time-consuming and with high costs. Since people have curved shapes on their bodies, they cannot come out as choreographers or directors have envisioned previously. Using DT takes advantage of this aspect, enabling the expansion or reduction depending on the scale or existing objects of the real stage. To make our system accessible to professional choreographers, we approach to use of intuitive interaction as they usually do in their real life. Similar to how professionals move tangible objects to plan the formation of the dancers, we use hand gestures with physical interaction for the operation of our system. Like drawing arrows on paper to represent movement, users can draw curves in 3D space to plan the trajectories of each dancer. Our system also enhances augmented collaboration by enabling multiple users to get connected for both online and offline collaboration in a DT-based XR environment.

2 RELATED WORK

According to the preliminary interviews of experienced choreographers we conducted, several choreographers physically draw directly on paper or board when making a trajectory and mark each dancer's next movement line with arrows. To supplement these more easily, several systems have been developed that can

simulate dancers' trajectories in advance because choreographers cannot predict all the movements of multiple frames.

To the best of our knowledge, there is no XR-based moving trajectory editing tool for choreographers so far. The most relevant case in the choreography field is kinds of motion capture for dance training in VR[3, 11, 21], enhancing movement system in AR[5] and with an interactive mirror[1] and for dancer education to give real-time visual guidance[4]. In addition, *Kirakosian et al.*[13] combine VR and AR for interactive dance training. However, we focused on the dancers' moving trajectory rather than choreography movement and devised the most efficient method of creating trajectories when a large number of dancers participated.

2.1 Choreography Production

There are several studies related to choreography production. One research [9] argues that creativity-support tools (CST) can play a valuable role in facilitating collaboration in dance-making. Also, flexibility and adaptability should be ensured in developing CST. At the core of our method, we ensure flexibility with our trajectory editing system. Also, we ensure adaptability by using a digital-twin-based VR environment so the created choreography can be adapted to various stages in the real world.

For the actual systems for choreography production, there exist some tools such as *Arrange Us*, *Choreo Note* and *Purple Pe*. These tools provide features for trajectory editing of each dancer but are mostly based on a 2D environment (*Arrange Us*, *Choreo Note*). Even for the 3D-based editing tools, there's no function exists to check the produced choreography from different view perspectives (*Purple Pe*). And because these tools represent dancers' movement lines by the linear interpolation of the positions in two frames, they cannot express the trajectory of each dancer accurately, especially for curved trajectories. Thus, the resulting movement usually becomes unnatural. Also, it cannot handle the case that two dancers move across each other since accurate editing of trajectories to avoid such overlaps is impossible. To resolve this problem, we propose a trajectory editing system that utilizes parametric curves so that it can be accurately edited to handle various situations.

2.2 Creation of moving line using Parametric Curves

The most intuitive way to create each dancer's moving trajectory is to draw lines by hand as the way existing choreographers do. However, one line consists of numerous points, and data has to be stored in units of those points. Because individual points corresponding to each frame in the timeline should have a curved connection by reflecting the user's intention as much as possible, we can use the parametric curve method.

In order for a user to create a movement line for each dancer on the system, it is necessary to draw a trajectory in the form of a curve. To represent the curve, parametric curves have been used for this purpose, which has long been studied in a way that create and manipulate curves in computer graphics. A parametric curve includes various control points, and there are two types to connect these points: 1) Interpolation and 2) Approximation. Curve points should be modified according to the user's intention or purpose according to each characteristic.

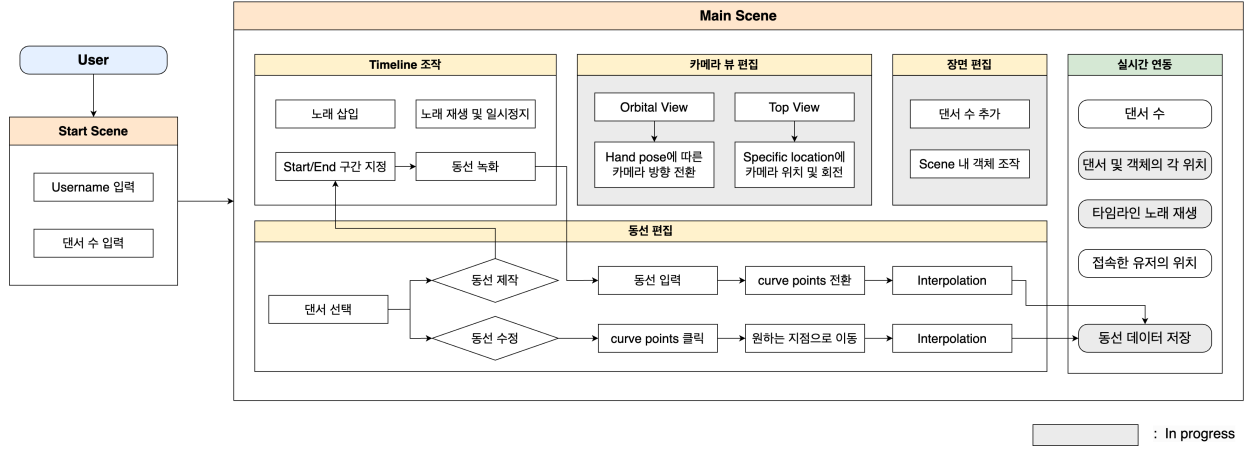


Figure 2: Pipeline overview of whole framework

Question	Answer	Our Approach
How do professionals usually plan the choreography?	plan formation first, use tangible objects, grab and move	Physically-based interaction - (A)
How do professionals plan the movement of each dancer?	transition between formations, draw lines	Trajectory drawing - (B)
Are there any difficulties regarding stages?	provided with blueprints, hard to deal with elevation differences and equipment	Digital twin-based VR environment - (C)
Do you use editing tools? If yes, what are the difficulties?	not much, way of use is too difficult, not intuitive	Intuitive hand gesture operation - (D)
What are the difficulties in conventional methods?	cannot consider multiple views, should be done with real dancers, hard to edit	Multiple camera views - (E)

Table 1: Results of survey on professionals. Each column indicates our question, answer, and our devised approach, respectively.

There are already a large number of motion interpolation approaches available to generate a continuous path from a discrete set of points. Conforming to [14], a third parametrization cubic curve basically had been used on the basis of heuristic analysis with a physical similarity. Within the Euclidean interpolation method based on this cubic interpolation, there are various types such as Hermite with more natural generalization and higher accuracy [6, 12], Catmull-Rom Spline with intersection-free [22] by controlling the parameter interval along the complexity of implementation is increasing. Among them, we have adopted the most basic cubic interpolation method [14].

Sometimes we do not usually have derivative data. To compensate this, there are C_1 (1st order), C_2 (2nd order) continuities in terms of tangent and curvature manipulation components individually for the local control and approximation in terms of continuity. In this case, there are also some approximation curves such as Bezier, B-Splines, Nurbs, and so on. The C_2 continuous composite cubic Bézier curve represents the control points while maintaining global control as much as possible. It was suggested to use 4 derivate points for estimation and can replace the Hermite interpolation [20]. On the other hand, a C_2 continuous cubic B-spline offers local control but only provides an approximation of the control points [10]. It is smoother than other interpolating curves and costs 3 times as much for curves. Therefore, we need to decide whether we prioritize global control or a closer approximation to the control points in choosing our approach.

3 PRELIMINARY SURVEY

To design a tool, we conducted a pre-interview with 2 professional choreographers for the survey of current choreography production trends. Both choreographers had more than 10 years of dance experience. Table 1 shows the survey result.

Physically-based Interaction. The professionals answered that for choreography production, they first plan the formation of dancers and then plan the trajectories of each dancer to smoothly connect those series of formations. To plan the formation of dancers, they answered that they usually grab and move tangible objects to get a sense of dancers standing in specific formation (Tab 1 (A)). From this response, we opted to use physically-based interaction, where users can grab and move the 3D dancer object with their hand interaction (Sec. 4.1) as they do in real-world situations.

Trajectory Drawing. After planning the formation of dancers, choreographers plan the movement of each dancer between the subsequent formations. Professionals answered that they usually draw a series of formations on a whiteboard or paper, and then draw lines for each dancer (Tab 1 (B)). From this answer, we decided to use trajectory drawing, where the user can draw the lines for each dancer to represent movements (Sec. 4.2).

Digital Twin. Professionals have mentioned that they are usually provided with the blueprints of the stages, but the real stages contain unexpected objects. This discrepancy between the real stage and provided blueprints leads to choreographers and dancers changing their planned movement and choreography right before their performance (Tab 1 (C)). To solve such an issue, we selected to use

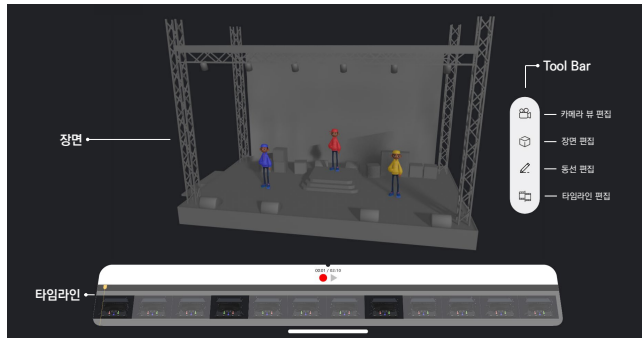


Figure 3: UI composition of the system

a digital twin-based VR environment (Sec. 4.4) so that they could check the interference of the stage objects from the very early stage of their work.

Hand Gesture Operation. The professionals answered that the currently available choreography editing tools are difficult to use and not intuitive (Tab 1 (D)). This is because the available tools heavily rely on touch or cursor interaction on mobile applications. Also, the tools do not contain capabilities for fine controls for dancers' movements, thus it takes much more amount of time to get the desired results. Thus we determined to use intuitive hand gestures for tool operation, provide a large capacity for drawing the trajectories, and edit the trajectory and timing information (Sec. 4.1, 4.2).

VR. Lastly, professionals have mentioned that conventional methods and tools are not capable of considering multiple views (Tab 1 (E)). Since it's hard to imagine especially the interference between dancers, most of the detailed planning should be done with real dancers. This means that every dancer should gather to check formations and choreography, which makes the process to be time-consuming and hard to edit. To ease the process and remove the needs of every dancer being gathered, we adopted to use of multiple camera view, by utilizing a VR environment (Sec. 4.1). In addition, by moving an actual auditorium environment to the virtual space, they can consider the perspective view of the audience. It saves the trouble of drawing a blueprint when there is a movement in the audience seats by referring to the whole planar figure of the venue.

4 METHODS

An overview of our method is shown in Fig. 2. Digital twins could be achieved with 3D spaces and technologies showing the environment using such AR/VR/MR head-mounted display (HMD) devices[8]. Our system was produced by Unity Engine, with a VR environment and Oculus Integration and users can enter the 3D areas with Meta Quest HMD.

In the 3D virtual space based on DT (Sec. 4.4) of real stages (Tab 1 (C)), users can control 1) camera view, 2) scene, 3) motion trajectory, and 4) timeline to perform choreography editing (Sec. 4.1). For seamless integration among dance parts, users collaborate via a constructed server environment (Sec. 4.3). Specifically, motion trajectory is described using parametric curves from the user sketched curves (Sec. 4.2). In the following sections, we will describe the details of each component.

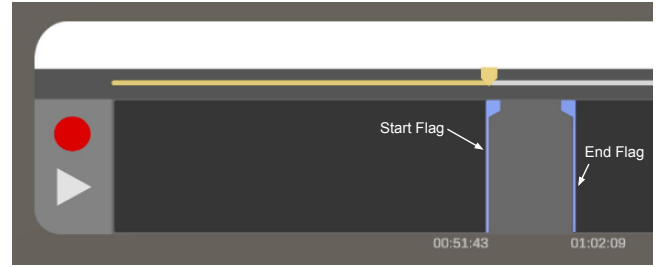


Figure 4: Start and end flag for specifying time range for the curve editing.

4.1 User Interface and User Experience

The user interface of our method is composed of 1) scene, 2) timeline, and 3) toolbox as described in Fig. 3. Scene is the place where users can interact with virtual objects and dancer avatars in 3D space, and see the result of choreography plans. Multiple users entering the system can communicate with each other about the planning of production progress by picking up avatars and placing them here and there freely using real-time voice chat. This enables users to collaborate simultaneously through the remote-based editing simulation system. Timeline is where users can insert and play the music, and adjust the timing of sections of dancers' trajectories by setting the start and end flags. The toolbox provides useful functions that are needed for users to edit some components for choreography production. Features of the toolbox will be described as follows.

Camera View Edit. Users can choose a specific camera view for the entire scene (Tab 1 (E)). Using the orbital view, users can walk around the digital twin stage to check choreography from various perspectives. With the pose detection function using Oculus Integration, the camera moves in the direction of the finger accordingly recognizing a set hand motion. The top view is optimal for checking the overall configuration of dancers fixing the camera at a specific location vertically over the stage.

Scene Edit. Depending on the situation, users can add more dancer avatars or virtual objects using the scene edit feature as they want. Based on Oculus hand tracking, users can easily grab and move these objects as they usually do in reality at their pleasure (Tab 1 (A), (D)).

Curve Edit. Users can create or edit the movement trajectory of each dancer avatar by simply sketching curves with distance hand grab interaction (Tab 1 (B)). The time range of the desired movement can be specified by the start and end flags as shown in Fig. 4. A parametric curve is used to describe these user-provided curves. Detailed methodology is explained in Sec. 4.2.

4.2 Trajectory Editing System

There are two options in the trajectory system, which are the creating mode and the editing mode. In creating mode, users can draw new trajectories for each dancer. In editing mode, users can simply modify existing trajectories by editing their control points. Each option will be explained in more detail in the following section.

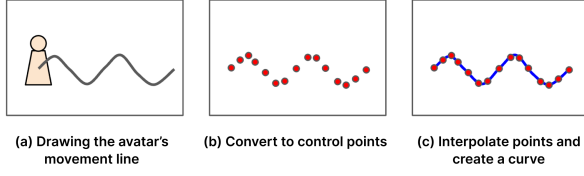


Figure 5: Interpolation method when the user creates a new trajectory of an avatar

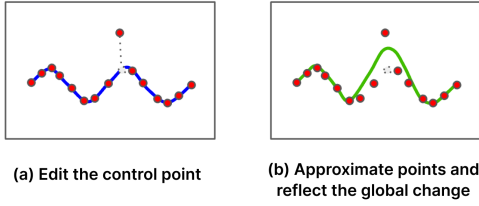


Figure 6: Approximation method when the user edits existing trajectory of an avatar

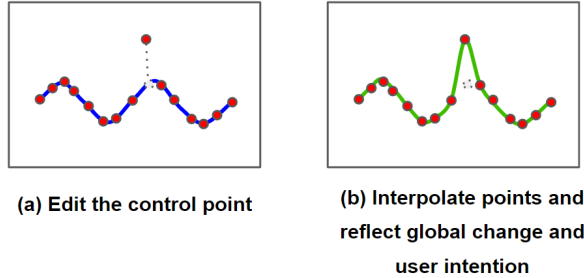


Figure 7: Interpolation method when the user edits existing trajectory of an avatar (Ours)

Trajectory Creation. If the user tries to draw the dancer's trajectory for the first time, enter the creation mode and select a specific avatar then draw the movement line as desired [See Fig. 5]. Control points are created according to the line drawn by the user, and parametric curves are formed in a way that connects them. To reflect the intention of the user as much as possible, the interpolation method is adopted. After the parametric curve automatically generates the interpolated curve based on these points, the movement of a specific avatar is created.

For more implementation details, users can produce the movement line of a specific dancer avatar for their desired timing. If the start and end times are designated as flags in the timeline [See Fig. 4], the time difference between the two flags is calculated. This time difference will be later utilized to estimate the number of final curve points. Then users grab the dancer avatar and move it to

draw the trajectory of their intentions (Fig.8(a)). To fulfill users' intentions, we visualize the object users' hands are grabbing (or hovering) by changing the color of the object from white to black. This has the effect of allowing the user to grab the desired object. While users are grabbing and moving the dancer avatar, our tool automatically tracks the position of the avatar, and spawns spheres on the tracked position are saved (Fig.8(b)). These spheres are considered as control points of the trajectory. Three fixed values are utilized while creating control points: 1) Record interval is the value that allows points to be created in fixed frequency, which makes the density of points vary depending on the speed of the user drawing. 2) Minimum distance is the value that constrains points not to be created in an unintended, unnecessary situation. 3) Maximum number of points is the value that constrains points not to be created infinitely. When the user releases the grabbed dancer avatar, spline interpolation is conducted with the coordinates of control points. In order to interpolate the real-valued function $f(x)$ at the $x + 1$ points when $x_0 < x_1 < \dots < x_n$, where the values of $f(x)$ are known, we can build a simple cubic spline polynomial as $S(x)$ referring to [7, 16].

The formula of cubic spline can be simply denoted as follows:

$$S_i(x) = a_i(x - x_i)^3 + b_i(x - x_i)^2 + c_i(x - x_i) + d_i \quad (1)$$

$$S_i(x_i) = f(x_i), S_i(x_{i+1}) = f(x_{i+1}) \quad (2)$$

Based on the above equations, x_i is i th control point and S_i is an equation of i th cubic spline, which can be denoted as a third-degree polynomial in Eq.1. From that representation, each control point can be equated in-spline continuity as part of the cubic line curve in Eq.2.

$$S_{i+1}(x_{i+1}) = S_i(x_{i+1}) \quad (3)$$

$$S'_{i+1}(x_{i+1}) = S'_i(x_{i+1}) \quad (4)$$

$$S''_{i+1}(x_{i+1}) = S''_i(x_{i+1}) \quad (5)$$

Each property contains more attributes of the cubic spline for the decision of connection of multiple curves. To combine two different curves into one curve, we have to ensure the inter-spline C_1 continuity for connection as you can see in Eq.3 while matching the end-point of the previous curve with the first point of the next curve. Even though the unification of those locations of two points, there should be a junction of two circular arcs for C_0 and C_1 continuity following the Eq.4 matching the tangent smoothness of both points. For more smoothness of curvature of C_2 continuity with those points, there should be an assurance based on Eq.5. The curve connecting several control points is derived by solving the above equations. The number of curve points is equal to the number of scene frames because we need the position of the dancer avatar at each frame to update the scene. We can estimate it by multiplying the time difference of the trajectory by a fixed fps rate (50 fps for our project). The time difference is calculated with those two-time flags designated at the start of the trajectory creation. Finally, we add the component in the trajectory object to save the property of the trajectory. In the component, the coordinates of curve points are saved in a type of array with a dancer index and two-time flags.

This component will be later referenced in trajectory edit mode to modify the existing trajectory.

After the creation of the trajectory, it (control points and curve points) is hidden temporarily for the visibility of the whole scene. The hidden trajectory is visualized later in edit mode. The movements of the dancer avatars synchronize with the music based on the created trajectory. Users can watch the avatar following trajectory by pressing the play button or moving the music time slider depicted in Fig. 4.

Trajectory Edit. If the movement data already exists and users want to modify the existing data, enter the editing mode then select a specific avatar and adjust the control points. Then, a new trajectory will be created with adjusted control points. At this moment, global changes are considered rather than local changes in the curve to tune the overall movement of the curve.

There are two simple approaches to reflect global changes in new trajectories. Adopting the approximation method rather than interpolation is the first approach to reflect global changes [See Fig.6]. Former studies on approximation curve [10, 20] have shown that approximation curve reflects more global changes to smooth the curve more naturally. The second approach to reflect global changes is to decrease the number of control points. We can reduce the number of control points by adjusting the fixed value mentioned earlier in the trajectory creation mode. More specifically, a large record interval and large minimum distance would result in the reduction of control points. If the number of control points decreases, the overall influence of each control point at the whole line increases, allowing for a better reflection of global change. However, both approaches could excessively alter users' intentions. First one might over-smooth the curve which can be described as distortion of intent at edit mode. Another one might lack the initial information to interpolate the first curve, which can be described as a distortion of intent in creation mode. In this study, a second approach is adopted while ensuring it does not excessively alter the user's intention [See Fig.7]. Reflection of global change and user intention are in a trade-off relationship. We have found a compromise by applying the most appropriate value to the fixed value, enabling us to adequately reflect both intent and global change.

For more implementation details, users can edit each existing trajectory individually. When users enter the edit mode, the trajectory corresponding to the current music time is visualized. Users can grab control points remotely, similarly to how the dancer avatar is grabbed in trajectory creation mode. After the user releases the grabbed control point, cubic spline interpolation is conducted with changed coordinates of control points (Fig.8(c)). At this moment, the component holding the property of trajectory is referenced to conduct spline interpolation. A modified trajectory is updated timely (Fig.8(d)). However, in an exceptional case where the end-points of the control points are edited, both adjacent trajectories are modified to ensure the continuity of the full trajectory.

Our method ensures and concentrates on users' ease of editing trajectory in two terms of convenience: First, using the parametric curve itself increases the convenience of editing trajectory. Users can edit continuous trajectories by modifying discrete control points. Consequently, drawing a parametric curve with few control points is better than initially saving a continuous curve in terms

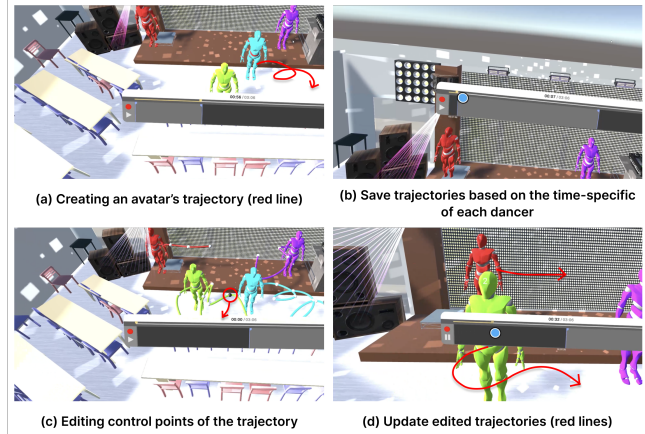


Figure 8: System functions that can be performed by the user for trajectory creation and editing of dancer avatars.

of editing trajectory. Moreover, saving control points in differing densities has an effect of a further increase in the convenience of editing. Control points are initially created in density that depends on drawing speed. We assume that draw speed has a correlation with the complexity of the curve, which means more control points are created in the complex section of the curve. Users can edit the trajectory separately: the complex section carefully, and the simple section roughly.

4.3 Server

One of the most important factors in digital twins as a virtual representation is the concurrent data simulation that applies to all multiple users. Integrating the diverse aspects of simulation technology, the same results should be reflected even in the different dimensions [19]. For the multi-level access operation [2], the server system should be arranged.

In order to implement an environment in which multiple users can access the VR simultaneously, we used Photon Unity Networking (PUN2) in Unity Engine. This makes network synchronization and remote process call (RPC) easy to use in Unity. In Fig.2, there are some components that should be updated for real-time interworking. Because all users who enter the system must see the same scene, 1) the number of dancers, 2) the position of each avatar and object, 3) the playing moment of a song on the timeline, and 4) the location of each connected user has to be matched. Those factors are updated from time to time and each user connected to the system can see the same results of them on the VR screen.

In this system, all users who access the server enter the Start Scene of Unity (Fig.9(a)), adjust the number of sliders of dancers, and enter the Main Scene. Users can also check each other's results and provide feedback while working on the choreography (Fig.9(b)). On the UI of Timeline, the section allocated by each worker is displayed (Fig.9(c)). This whole function enables real-time collaboration.

Usually, in the process of choreography trajectory production, several choreographers are in charge of one choreography. That's why it would be better for all choreographers to participate remotely in the digital twin and collaborate in real-time while moving their

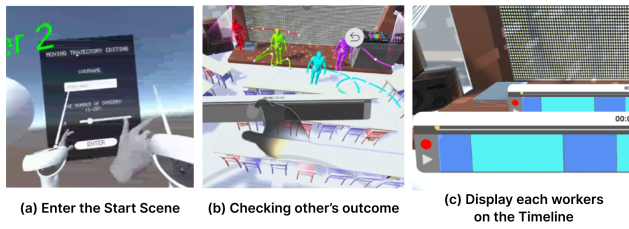


Figure 9: System functions that can enhance real-time collaboration in a digital twin environment.

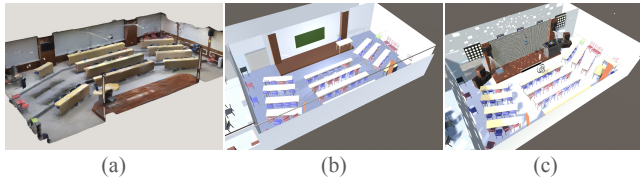


Figure 10: DT construction process. (a) From the raw scanned data, (b) we registered it with new meshes, (c) and then augmented the environment as a stage with virtual objects.

dancer avatars in order to be well aware of the dance concept and continue the mood of choreography regularly. If they collaborate at the same time, they can interact with each other and make and compare various proposals, and they can expect better results by choosing the optimal method.

4.4 Digital Twin

We build our system upon a digital twin (DT) environment. Using DT environment for choreography editing takes advantages in the following two aspects: 1) Users can edit and check the editing results on a real-world scale, and 2) Users can reflect the real-world environment or objects of the stage into their choreography plans. To construct a DT environment, we follow the two steps of the process, which are a) Registration, and b) Augmentation. Fig. 10 shows the entire process of DT construction with visual examples. In the following paragraphs, we explain each process in detail.

Registration. The convert real space into DT, the first step is to scan the space information. To this end, we scan the space with the built-in stereo camera of HoloLens2. However the raw scanned point cloud data are sparse, so it usually contains holes which makes it hard to be augmented with the virtual objects. To resolve such issues, we create a dense DT space with new meshes and objects, referring to the scale and the structure of the real space that is scanned. With this registration process, the constructed DT space can be augmented with virtual objects to make stages for the performance, while maintaining the scale of the real space.

Augmentation. After the registration process, we augment the space with virtual objects so that it can represent the stage environment. We place a speaker, lighting, light screen, and the flammars as its real stage. Using this information, choreographers can plan the dancers' moving trajectory with no interference with the stage

objects. Also, they can check how it would be delivered to the audience by placing the camera in a specific location.

5 CONCLUSION

Through our system, choreographers can make the moving trajectory of a number of individual dancers effectively in the VR environment. This reduces their choreography production load a lot, and it is expected that various performances can be produced on a larger scale and accelerated by improving usability and convenience in the dancing field. In addition, since the remote system in VR is utilized, choreographers can actually collaborate in real-time without meeting face-to-face, reducing spatio-temporal costs.

For future work, it seems there are several factors that can be complemented by our systems. If the human factor or data-driven method, such as the data of speed or the movement of a real person is applied, it will be possible to simulate it more realistically. In order to implement realistic pedestrian trajectories, the reinforcement learning model has been used to reflect user-defined goals and implement object or crowd avoidance with surrounding environmental data [15, 17, 18]. Especially in the case of mega-crew performances, since there are many dancers and people should not be covered or bumped together, it would be good to automatically consider those points.

If the dancing-motion retargeting is also assigned for each dancer's avatar, it will be more useful than only applying the moving trajectory. If the system is distributed to dancers beyond the trajectory-editing tool later, it is easy for them to learn the overall choreography composition in advance and the hassle of having to acquire new route changes depending on the actual size of the concert hall will be eliminated.

Furthermore, a simulation of the dancing performance will be possible beforehand by utilizing the movement data stored at other venues. It means that if there is 3D spatial-scanned data from other performance halls, not only the stored movement data can be applied in the VR, but also you can actually go to another place and then augment the data on the ground.

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